HW 1 10 points

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What tracks did you choose? Please check all tracks you chose.

- ✓ Genius Track ( )
- ✓ Transparency Track (v)
- ✓ Consistency Track (v)

1. (Bonus 1 point) Is the average-case running time of insertion sort linear, quadratic, or of some other order? Prove your answer completely.

Quadratic.

$k \to$ already sorted part of size $k-1$.

$$\left.\begin{array}{l} 1 \to 0 \\ 2 \to 1 \\ \vdots \\ k \to k-1 \end{array}\right\} \quad \text{average} : \frac{(k-1)+(k-2)+\cdots+1+0}{k}$$

$$= \frac{\frac{\cancel{k}(k-1)}{2}}{\cancel{k}} = \frac{k-1}{2} \approx \frac{k}{2}$$

$$T(n) = \sum_{k=1}^{n} \frac{k}{2} = \frac{1}{2}\sum_{k=1}^{n} k = \frac{1}{2} \cdot \frac{n(n+1)}{2}$$

$$= \frac{n^2+1}{4}$$

$$\therefore T(n) \approx \frac{n^2}{4} = \Theta(n^2)$$

Transparency

i. I asked ChatGPT that what is the meaning of the average-case running time?

ii. https://chatgpt.com/share/69b7a108-564c-8012-a673-de7b2c84bf5d

2. (2 points) Find the simplest form of $\Theta(10^{10}n^2 + 10^5 n + 10^9)$.

$\Theta(n^2)$

When n becomes infinitely large,

1. $10^{10}n^2$ grows by n^2
2. $10^5 n$ grows by n
3. 10^9 is constant.

$n^2 > n > 1$

So, $\Theta(10^{10}n^2 + 10^5 n + 10^9) = \Theta(n^2)$

Transparency

i. I used AI. I already know about the answer. But I didn't know the exact process of how the answer came out. So I asked the reason why the simplest form becomes the highest growth term.

ii. https://chatgpt.com/c/69b59b1a-1a20-8324-ac01-9484f19ac85f

3. (4 points) Watch the video "Growth Rates of Functions and L'Hospital's Rule" (https://youtu.be/6IjVGfbhe9w) and answer the following question.

Where does $O(n^{\frac{1}{100}})$ belong? Prove your answer completely.

$$O(1) \subset ① \subset O(\lg n) \subset ② \subset O(\sqrt{n}) \subset ③ \subset O(n) \subset ④ \subset O(n \log n)$$

②

$\lim_{n\to\infty} \frac{\lg n}{n^{\frac{1}{100}}} = \frac{\infty}{\infty}$ ← We can use L'Hospital's Rule

$\lg n = \log_2 n = \frac{\ln n}{\ln 2} = \frac{1}{\ln 2} \times \ln n = \ln n$ asymptotically same.

$f(n) = \ln n \rightarrow f'(n) = \frac{1}{n}$

$g(n) = n^{\frac{1}{100}} \rightarrow g'(n) = \frac{1}{100} \times n^{\frac{1}{100}-1} = \frac{1}{100} n^{-\frac{99}{100}}$

We will use $\frac{f'(n)}{g'(n)} \rightarrow \lim_{n\to\infty} \frac{\frac{1}{n}}{\frac{1}{100} n^{-\frac{99}{100}}}$

$= \lim_{n\to\infty} \frac{100 \cdot n^{-1}}{n^{-\frac{99}{100}}}$

$n^{-\frac{100}{100}} \div n^{-\frac{99}{100}}$
$= n^{-\frac{100}{100}-(-\frac{99}{100})}$
$= n^{-\frac{1}{100}}$

$\lim_{n\to\infty} \frac{100}{n^{\frac{1}{100}}} = \lim_{n\to\infty} 100 \cdot n^{-\frac{1}{100}}$

$\rightarrow 0$

$\therefore O(\lg n) = O(n^{\frac{1}{100}})$
$\Rightarrow O(\lg n) \subset O(n^{\frac{1}{100}})$

Then, $\lim_{n\to\infty} \frac{n^{\frac{1}{100}}}{\sqrt{n}}$

$= \lim_{n\to\infty} (n^{\frac{1}{100}} \div n^{\frac{1}{2}}) = \lim_{n\to\infty} n^{\frac{1}{100}-\frac{50}{100}} = \lim_{n\to\infty} n^{-\frac{49}{100}}$

$= \lim_{n\to\infty} \frac{1}{n^{\frac{49}{100}}} \approx 0$ $\therefore O(n^{\frac{1}{100}}) = O(\sqrt{n})$

$\Rightarrow O(n^{\frac{1}{100}}) \subset O(\sqrt{n})$

So, $O(\lg n) \subset O(n^{\frac{1}{100}}) \subset O(\sqrt{n})$

The answer is ②

Transparency

i. I used AI to reconfirm the concept of L'Hôpital's Rule and study how to differentiate logarithms.

ii. https://chatgpt.com/share/69b7af40-28f4-8012-ab57-703f5eb0c68b

4. (Bonus 1 point) Which one is correct? Prove your answer completely.

$O(n2^n) \subset O(n!)$ or $O(n!) \subset O(n2^n)$

$$O(n2^n) \subset O(n!)$$

To prove $O(n2^n) \subset O(n!)$, $n2^n \subset O(n!)$

If $\lim_{n\to\infty} \frac{n\cdot 2^n}{n!} = 0$, then $n2^n \subset O(n!)$.

I will use ratio method because of factorial.

$a_n = \frac{n2^n}{n!}$, $a_{n+1} = \frac{(n+1)\cdot 2^{n+1}}{(n+1)!}$ ↳ $(n+1)! = (n+1)(n!)$

$$\frac{a_{n+1}}{a_n} = \frac{\frac{(n+1)\cdot 2^{n+1}}{(n+1)!}}{\frac{n\cdot 2^n}{n!}} = \frac{(n+1)\cdot 2^{n+1}\cdot n!}{(n+1)!\cdot n\cdot 2^n} = \frac{\cancel{(n+1)\cdot n!}\cdot 2^{n+1}}{\cancel{(n+1)!}\cdot n\cdot 2^n}$$

$$= \frac{2^{n+1}}{n\cdot 2^n} = \frac{2\cdot \cancel{2^n}}{n\cdot \cancel{2^n}} = \frac{2}{n}$$

$\lim_{n\to\infty} \frac{2}{n} = 0$ $\therefore a_n \to 0 \Rightarrow \lim_{n\to\infty} \frac{n\cdot 2^n}{n!} = 0$

So, $n^{2n} = O(n!) \Rightarrow O(n2^n) \subset O(n!)$

Transparency

i. I studied how to perform limit calculations when there is a factorial using AI.

ii. https://chatgpt.com/share/69b6bba3-b948-8012-8a80-76974690905d

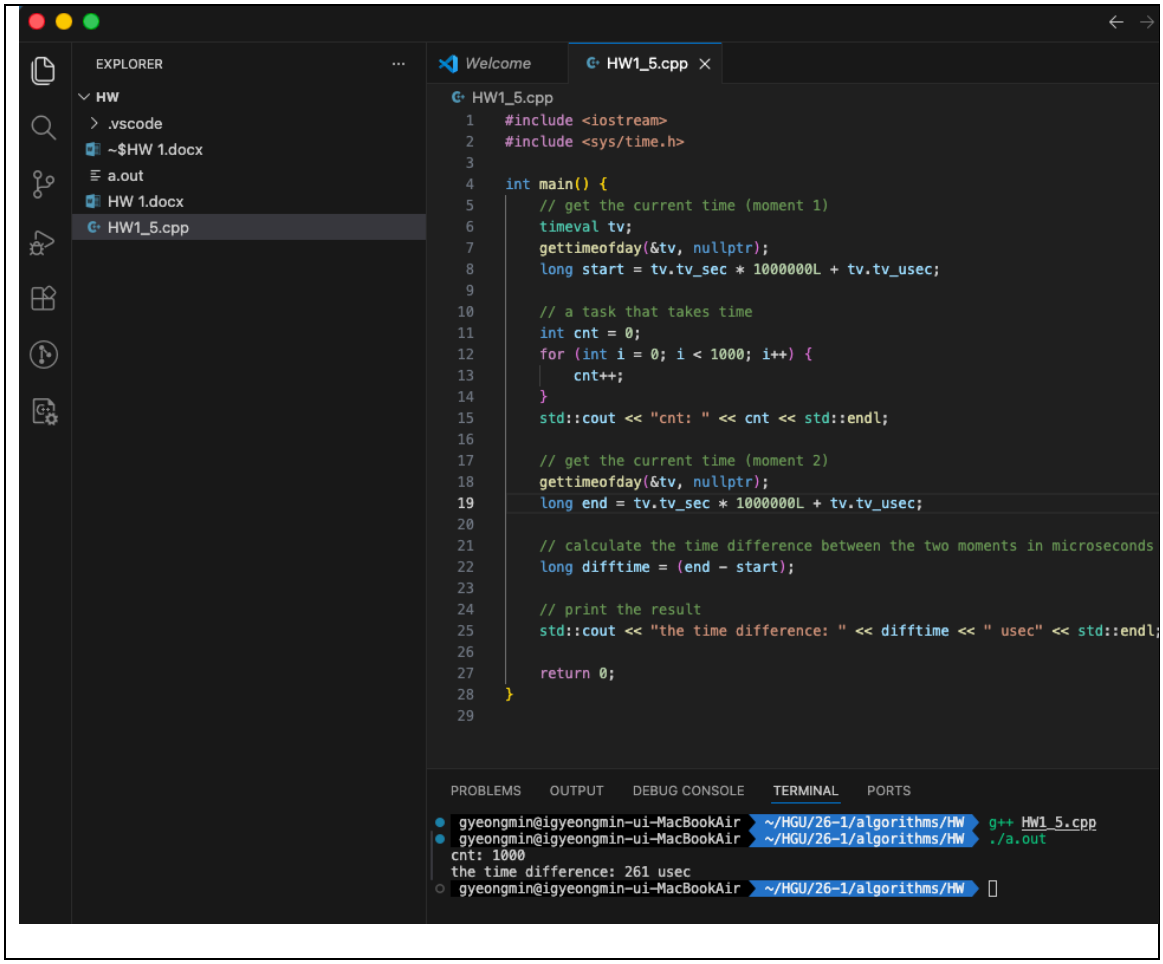
5. (4 points) Run the following c++ code and get the result.
   Screen capture your result and include your result image(s).

```
#include <iostream>
#include <sys/time.h>

int main() {
    // get the current time (moment 1)
    timeval tv;
    gettimeofday(&tv, nullptr);
    long start = tv.tv_sec * 1000000L + tv.tv_usec;

    // a task that takes time
    int cnt = 0;
    for (int i = 0; i < 1000; i++) {
        cnt++;
    }
    std::cout << "cnt: " << cnt << std::endl;

    // get the current time (moment 2)
    gettimeofday(&tv, nullptr);
    long end = tv.tv_sec * 1000000L + tv.tv_usec;

    // calculate the time difference between the two moments in
microseconds
    long difftime = (end - start);

    // print the result
    std::cout << "the time difference: " << difftime << " usec" <<
std::endl;

    return 0;
}
```

cnt: 1000

The time difference: 261 usec

HW 1 10 points



For Genius Track students

Write your reasons for challenging the bonus problem(s) and your thoughts here.

Transparency Track

Describe what you realized you did not fully understand through this assignment. here.

I did not know the exact meaning of average-case running time, nor did I fully understand why the highest growth term is sought when finding the simplest form. Through this assignment, I learned what L'Hôpital's rule is and when to use it, and I also learned how to calculate limits when factorials are involved.

Appendix

How to compile and run your c++ code:

1. You may use any tool if you already have.
2. Or go to Google Colab and write down the code in this way.
   A. Write your code like below in the first cell but you must include `%%writefile program.cpp` at the top of the code. Then, run the cell.

```
%%writefile program.cpp
#include <iostream>
#include <sys/time.h>

int main() {
    // get the current time (moment 1)
    timeval tv;
    gettimeofday(&tv, nullptr);
    long start = tv.tv_sec * 1000000L + tv.tv_usec;

    // a task that takes time
    int cnt = 0;
    for (int i = 0; i < 1000; i++) {
        cnt++;
    }
    std::cout << "cnt: " << cnt << std::endl;

    // get the current time (moment 2)
    gettimeofday(&tv, nullptr);
    long end = tv.tv_sec * 1000000L + tv.tv_usec;

    // calculate the time difference between the two moments in
microseconds
    long difftime = (end - start);

    // print the result
    std::cout << "the time difference: " << difftime << " usec"
<< std::endl;

    return 0;
}
```

   B. Write the following code in the second cell. Then, run the cell.

```
%%shell
g++ program.cpp -o execute
./execute
```